

WEEK 4: DATA INTEGRATION AND FEATURE ENGINEERING

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Project: Heavy Supplier, Inventory & Warehouse Analytics

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Objective:

To understand the concepts and techniques involved in data integration and feature engineering and how they can be applied to the CadetX project.

Data Integration

Data integration is the process of combining and harmonizing data from multiple sources into a unified, coherent format that can be put to use for various analytical, operational and decision-making purposes.

Key Concepts (Researched):

- ❖ **Data Modelling:** the process of structuring your data and defining the relationships between different tables so that Power BI can understand how your data is connected.
- ❖ **Fact & Dimension** Fact tables typically contain measurable business events or values and foreign keys that connect them to dimension tables. **While**, Dimension table on the other hand, primarily contain primary key & descriptive attributes.
- ❖ **Granularity & Uniqueness:** Granularity refers to the level of detail represented by each row in a table. **GRAIN** = What one row represents.
- ❖ **Keys & Relationships**
- ❖ **Cardinality:** Cardinality in a relationship defines how many rows in one table can be related to rows in another table.
- ❖ **Star Schema:** It is a model, where the fact table sits at the center with the dimension table surrounding it.
- ❖ **SQL JOINS:** Inner Join, Left Join, Right Join, & Full Outer Join.

Application to the CadetX Dataset

- ❖ Identifying what each table represents.
- ❖ Understanding the grain of each table
- ❖ Identifying possible keys
- ❖ Determining relationships between tables
- ❖ Planning how the tables can be integrated for analysis

Feature Engineering

Feature Engineering means creating new useful columns from the existing columns for easy analysis.

Key Learnings

- ❖ Understanding the grain of each table is important before establishing relationships.
- ❖ Different JOINS produce different results and should be selected based on the analytical requirement.
- ❖ Proper data modelling helps organize related tables for accurate analysis.

Next Steps

- ❖ Continue integrating the cleaned datasets
- ❖ Develop the data dictionary
- ❖ Validate table relationships
- ❖ Implement relevant engineered features
- ❖ Prepare the integrated data for analysis and visualization in Power BI.